Hasan Bukhari

Software Engineering • Systems • Backend & Applications

+1 601 913 4692 | hasan.bukhari25@gmail.com | LinkedIn | GitHub | Portfolio

Education

University of Southern Mississippi — **Expected Dec 2028**
B.S. in Computer and Information Sciences; Minor in Biological Sciences — *Hattiesburg, MS*

GPA: 4.00 • President's List • Academic Excellence Scholar

Beaconhouse College Program — **May 2024**
*Cambridge A Levels: 4 A*s in Biology, Chemistry, Mathematics, and Physics* — *Karachi, Pakistan*

Software Engineering Experience

Data Analyst Intern, Health, Safety & Environment — **Jul 2026 – Aug 2026**
Engro Technical Services — *Karachi, Pakistan*

- Engineered Python, FastAPI, and Power Query pipelines for risk scoring, scheduled reporting, and automated change monitoring across approximately 1,750 controlled engineering and HSE files.
- Built Power BI data models, DAX measures, and executive analytics over 130,000+ incident, near-miss, inspection, audit, waiver, and corrective-action records.

Student Worker, Operations Technology & Event Support — **Jun 2024 – Present**
Thad Cochran Center / Union Operations, University of Southern Mississippi — *Hattiesburg, MS*

- Built SetItUp and UnionInventoryPro to digitize room requests, event layouts, equipment tracking, and recurring inventory audits for university operations.

Undergraduate Researcher, Ichthyology Lab — **Spring 2026 – Present**
University of Southern Mississippi — *Hattiesburg, MS*

- Build reproducible Python workflows and chromosome-correspondence matrices to analyze chromosome-scale genome assemblies and comparative-genomics evidence.

Software Engineering Intern — **Aug 2023 – Feb 2024**
Shahid I.T. Consultants — *Karachi, Pakistan*

- Automated recurring record-update workflows in Python, reducing manual processing by 30%; supported integration testing, deployment documentation, and client application maintenance.

Selected Software Projects

AEGIS — Controlled Document Intelligence Platform | Python, FastAPI, React, TypeScript, Rust/Tauri | GitHub — **2026**

- Built a local-first Windows desktop platform for deterministic file monitoring, retained-version diffing, governance workflows, searchable analytics, SHA-256 integrity verification, and audit history.
- Engineered a FastAPI backend, React/TypeScript UI, Tauri desktop shell, regression/adversarial tests, database migrations, release QA, and native Windows packaging.

HEARTLINE — Native Android Lyric Companion | Kotlin, Jetpack Compose, Room, Android Services | GitHub — **2026**

- Built a native Android application with media-session detection, synchronized playback state, offline persistence, resilient service recovery, on-device image rendering, automated tests, lint, and CI builds.

To Ash Again — 2D Narrative Platformer | Python, Pygame, OOP | GitHub — **2025**

- Developed a four-act standalone game over two months with modular player/enemy systems, alternate-character mechanics, inventory and powers, progression, save state, cutscenes, and a multi-phase boss.

WCRE — Windows Checkpoint/Restore Engine | Rust, Win32, Windows PSS, x64 | GitHub — **2026**

- Developing an experimental Windows x64 checkpoint/restart engine with process inspection, PSS-backed checkpoint capture, persistent checkpoint images, restore planning, and controlled virtual-address reconstruction.

Technical Skills

Languages: Python, Rust, C++, Kotlin, Dart, JavaScript/TypeScript, SQL
Applications & Backend: FastAPI, Flask, REST APIs, React, Android, Jetpack Compose, Flutter, Room, Tauri
Systems & Data: Win32 API, Windows PSS, filesystem monitoring, pandas, NumPy, NetworkX, Power BI, DAX, Power Query
Tools & Infrastructure: Git/GitHub, GitHub Actions, Linux/WSL, Cargo, Gradle, PlatformIO, CI/CD, native Windows packaging

Leadership & Honors

President & Founder, Student Poets Association and Creative Writing Club, USM — Lead arts programming, public showcases, open mics, and organizational operations.

Selected Honors: Outstanding Cambridge Learner Award; ITC 371 Best Project & Best Assistant Project Manager; UGS Best Art as Research Project (2026); Best Arts Showcase for Poetry (2025, 2026); President's List.